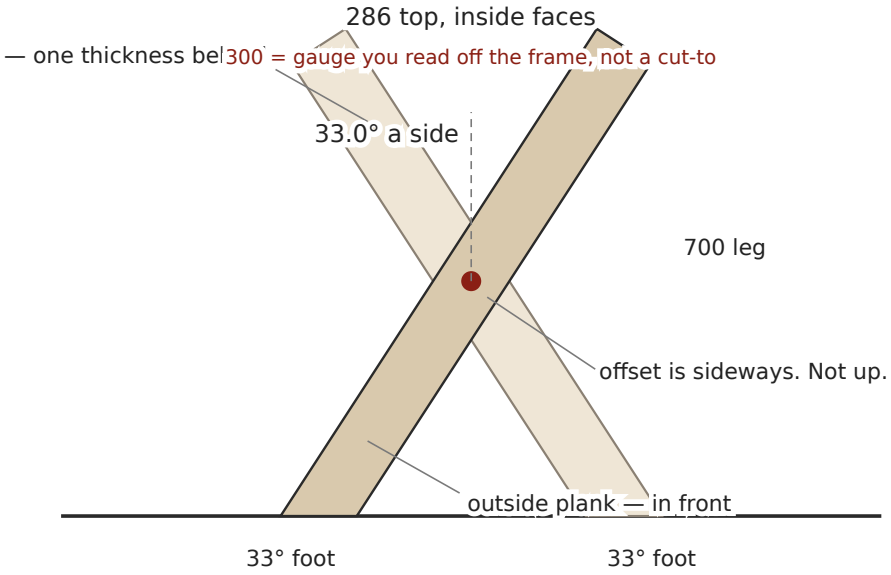
X-Leg Herb Trough

tutorial · v10

- Two X-leg frames carry a **900** trough.
- Legs **4×700**. Cross planks **8×900**. Planks **80 wide, 23 thick**.
- **Gauge 300**, inside corner to inside corner.
- Read the gauge off the frame. Never cut to it.
- Legs do not notch. One plank in front, one behind, **one thickness apart**.
- The ends are **scribed** off an offcut. Not a number.
- One trim only: **one plank thickness** off the second floor plank.
- Shown at 900 long, 4 planks deep. All in mm.

X JOINT - stage 1



The joint decides the box.

- Two 700 legs, crossed. **Both feet reach the floor.**
- The inside plank sits one thickness behind the outside plank.
- That offset is **sideways**, out of the page. Nothing rides up.
- Two frames. **Mirror pair:** the outside plank swaps sides.
- 35 mm part-threaded: through the near plank, bites the far one.
- Span between the inner faces: **286**. Worked out, not measured.
- **2 × (350·sin33° − 40÷cos33°) = 286**
- **300** is the gauge on the standing frame. Not a cut-to.
- Foot flush on the clamped plank, top at 300. Frame stands true. Both frames match.

CUT LIST

PART	QTY	LENGTH	WIDTH	THICK	NOTE
Leg	4	700	80	23	one end cut 33°, all four one setting
Cross plank (wall)	8	900	80	23	4 front, 4 back
Side piece	6	scribed	80	23	3 per end, marked off an offcut
- of those, the bottom pair	2	900	80	23	first plank's inner long edge cut 22.5° into the V; second trimmed one plank thickness

BUY IT IN YOUR YARD'S STOCK

- **Timber: 6 lengths of 2 m**, 80 wide × 23 thick.
- Legs: 4×700 = 2,800. **2 lengths.** Two legs per length.
- Walls: 8×900 = 7,200. **4 lengths.** Two per length, 200 off each.
- Offcuts left: two 600s, four 200s.
- **Watch the sides.** Six side pieces come off the two 600s. Longest first.
- Needs more than 600? Buy a 7th length.
- **The trim is 23.** One plank thickness. Your number, same rule.
- **Buying 90 wide instead?** Two numbers change. Nothing else does.
- Crossing mark: **45 across**. Half of 90, not 40.
- Top span between leg inner faces: **274**. Same formula, 45 in place of 40.
- 700, 900, 33°, 22.5°, the thickness trim: all the same.
- The drawing here is at 80 wide.
- Untreated pine or hardwood. Untreated: it is a food bed.
- **35 mm part-threaded screws** for the crossing and the side courses.
- A **6 mm or 10 mm bolt** through each crossing if you want it to never move.
- Stainless or coated only. Wood oil (Woodoc) outside and on the feet.

The rules read this first

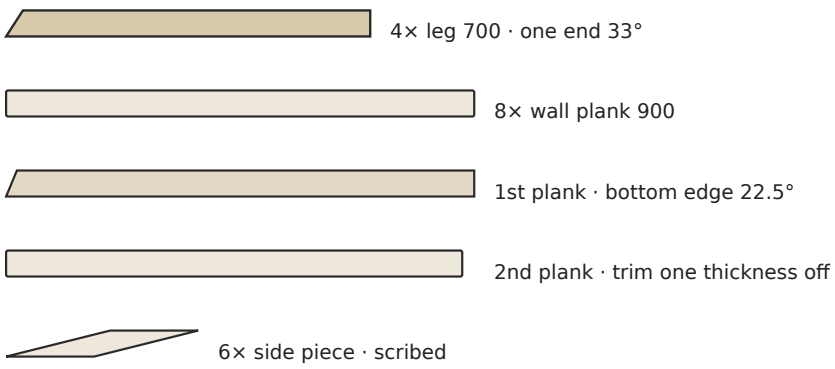
Two rules that decide this build

- **One saw setting for all four feet.** Cut every 33° end off the same stop. Cut one foot on its own and the box rocks.
- **Lock the crossing.** A single screw lets the X scissor. Add 3-4 screws, or bolt it.

Locked in from your build photos (14 Oct)

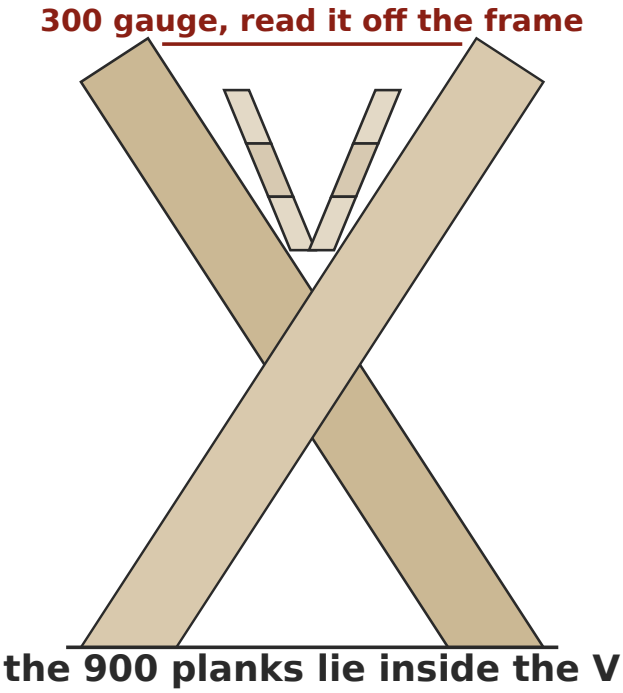
- **Top = 300, inside corner to inside corner.** Read it off the standing frame - it is the check, never a cut-to. Every X the same.
- **Feet cut at 33°** so they sit flat. Method: clamp a plank, leg flush, one centre screw into the middle of the plank.
- **The two legs never notch.** They lie parallel, one thickness apart, on the same centre line. Both feet therefore reach the floor. That gives an inside plank and an outside plank.
- **Spacer blocks at the shoulder.** Small blocks fill the gap between the angled leg head and the underside of the first planks. Drawn from the ~60 and ~75 mm marks in your shots.
- **One wall flush, one proud.** The back planks run **flush to the outside leg**; the front planks stand **one plank thickness proud** (23 mm for 23 mm stock), as in your shots.

EVERY CUT, DRAWN



Two angles hold the whole thing. **33°** on the leg. **22.5°** in the V. The rest square.

Where the planks sit end view



End view: the planks sit inside the V.

- Not a tray on top. It drops down inside the X.
- Both sides lean in at **22.5°** to a narrow bottom.
- Leg tops stand proud above the rim.
- Lowest plank: bottom edge cut at 22.5°, so it sits level.
- Next plank lands one thickness higher. **Trim one thickness off its long edge.** Tops come level.

The two angles that fool everyone

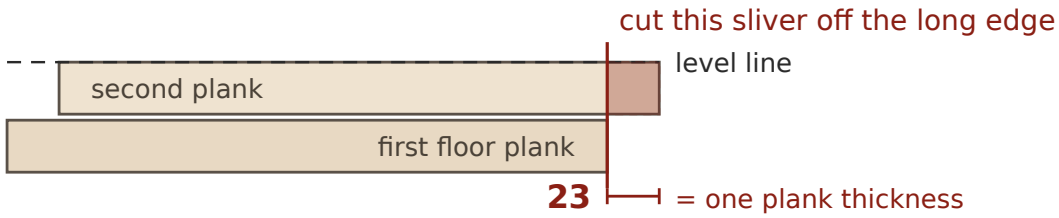
- 33° is the **leg** - the foot cut and the lean, one saw setting. Nothing else is 33°.
- 22.5° is the **V** - how far each side leans in, and the cut on the lowest plank's bottom edge to match it. A chosen call: it beds the plank into the V, it is **not** derived from the 33°.
- Each side above that is **scribed** off an offcut stood in the V, not set to a number.

The trim, up close

the one lengthwise cut

THE ONLY LENGTHWISE TRIM

23 is one plank thickness. 19 stock: trim 19.



How to get it level

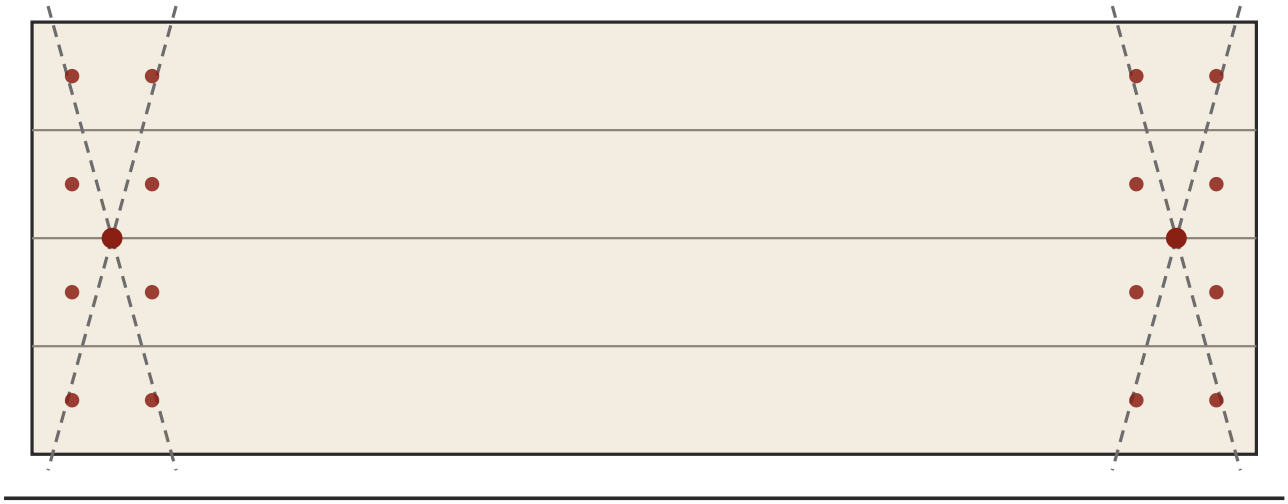
- First floor plank beds into the V on its 22.5° edge.
- Second plank lands **one thickness higher**. That is the gap.
- Trim **one plank thickness** off its long edge. The tops then sit level.
- 23 is your stock. 19 stock: trim 19. The thickness is the rule, not the number.
- Trim it before you screw. Screwed down and you are planing a fixed course.

Where the screws go

the count

SCREW MAP · front

dashed lines = the X frame, hidden behind the wall planks



WHERE	SCREWS	NOTE
Crossing, both frames	**8**	4 around the crossing, or 1 bolt each. One screw lets the X scissor.
Wall planks to legs	**32**	2 per plank end. 4 courses × 2 sides × 2 ends = 16 ends.
Spacer blocks at the shoulders	**8**	1 each. The block fills the gap the angled leg head leaves.
End pieces	**12**	3 pieces per end, 2 screws each. Glue as well.
Total	**60**	**A 200 box of 35 mm covers it.**

Two screw rules

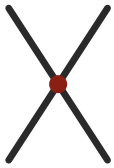
- **35 mm part-threaded.** It travels one plank thickness, then bites solid wood. Fully threaded pulls apart instead of clamping.
- **Pilot drill first.** 3 mm into pine, 3.5 mm into hardwood. No pilot, no second attempt.
- **Stainless or coated only.** It is a food bed.
- **Never screw into end grain** and expect it to hold. The spacer block is there so you do not have to.

The build in four stages

picture sequence

Four stages, in order. Do not start the sides until the walls are screwed down and level.

1

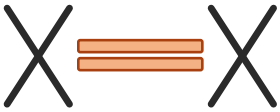


33° feet · screw the centre

The X legs.

- Cut one end of each 700 leg at 33°.
- Upper leg on top of the lower. Centre 350, 40 across.
- One screw through the middle. Make two, mirrored.
- Both sets at the same 300 top gauge.

2



first two in the V

Stand them, start the floor.

- Two X sets parallel, set to the 300 gauge.
- First 900: cut one long edge at **22.5°**, inner edge.
- Second 900 lands one thickness higher.
- Trim one thickness off its long edge.

3



4 front + 4 back

The rest of the 900s.

- 4 planks front, 4 back.
- Flush with the left and right X.
- Back sits one thickness proud.
- Screw down front and back.

4



sides off an offcut

The sides.

- Hold an offcut in the V. Mark the angle.
- Cut the first side piece. Fit it.
- Next two: mark off the one below. Cut, glue, screw on top.
- Mirror the other end.

Stage by stage 1 – 2

#	STAGE	WHY
1	**The X legs.** • Four legs, 700 long. Cut **one end of each** at 33° - all four off the **same saw setting**. • Find the centre of each leg: 350 along, 40 across (half of 80). • Clamp a straight plank to the table. Press the 33° end flush against it. • Lay the second leg alongside the first, **parallel, one thickness to the side**, to make the X. Pilot hole at the centre. • Drive a 35 mm part-threaded screw through the near plank so it bites the far one — one thickness of travel. • Then 3-4 more screws, or swap to a 6 mm or 10 mm bolt through the crossing. • Set the standing frame so the **top gauges 300, inside corner to inside corner** - read it, don't cut to it. Make the second set the same.	• One saw setting for all four feet = feet that sit level. Cut them one at a time and they fight. • The clamped plank is the reference - it stops the X leaning while you screw. • The legs do not notch: parallel, one thickness apart, both feet down. That gives an inside leg and an outside leg. • A single screw lets the X scissor. Extra screws or a bolt lock the crossing. • 300 top gauge on both sets = the box sits level and the walls line up.

> **Check before stage 2:** feet sit flat, both frames the same 286 inside gap. Do not move on until it is true.

2	**Stand them and lay the first two planks.** • Stand the two X sets parallel, on the ground. • Lay the first two 900 planks in the V, one front, one back. • Cut **one long edge of the first plank at 22.5°** - the inner edge where it meets the far leg - so it beds into the V. • The second plank lands **one plank thickness** higher, so **trim one plank thickness off** its long edge - 23 mm for 23 mm stock - and both tops come level.	• Both X sets parallel = the box goes up straight. • The 22.5° on the inner edge is what lets the plank bed into the V instead of rocking on the leg corner. • Level first planks = level box. This is the one that sets it.

> **Check before stage 3:** both starter planks snug in the V and level with each other. Do not move on until it is true.

Stage by stage 3 – 4

#	STAGE	WHY
3	**Lay the rest of the 900s.** • Carry on stacking: 4 planks on the front, 4 on the back, one on top of the other. • All flush with the left and right X planks. • The back stacks one plank thickness proud (23 mm here), behind the outside leg. • Both sides finish at the same height. • Screw the stack down, front and back, into the X frames.	• Flush to the X = straight walls, no stepping. • The back offset clears the leg, so both walls meet the frame the same way. • Equal height both sides = a level rim.

> **Check before stage 4:** both long sides finish level before you close the ends. Do not move on until it is true.

#	STAGE	WHY
4	**The sides, scribed off an offcut.** • Hold an offcut horizontal in the bottom of the V. Draw a line from inside to mark the angle of the first side piece. • Cut to the line, fit it in the V. • Hold a longer offcut above the first, mark off it, cut the second, glue and screw it on top of the first. • Third the same, on top of the second. • Do the same for the opposite end.	• The V is not a set angle you can measure off a drawing - you scribe each piece off the one below. • Glue **and** screw: end grain is weak, the glue takes the load, the screws hold it while it sets. • Three pieces fill the end from the floor to the rim.

> **Final check:** all six end pieces fit inside the V, three at each end. Do not move on until it is true.